

De Novo Antibiotic Candidate Screening

Generative LSTM Pipeline — Final Shortlist Report

Generated 2026-07-14

Generative Model Performance Summary

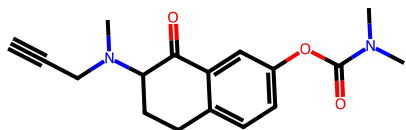
Trained for 54 epochs (early stopping on validation loss). Final token-level next-character accuracy: 0.914 (train), 0.866 (validation). This measures how well the model predicts the next SMILES character during training — it reflects syntax learning, not molecular or biological quality.

Temperature	Validity	Uniqueness	Novelty
0.5	0.920	0.924	0.370
0.7	0.700	0.986	0.486
0.8	0.680	0.971	0.632
1.0	0.480	1.000	0.729

What this report does NOT establish: No biological activity predictor (e.g. an RF model on assay data) has been applied in this pipeline. Ranking below is by **CNS MPO score (5-parameter approximation)** — a structural drug-likeness heuristic, not a prediction of antibacterial activity. These are computational hypotheses; experimental validation (activity assay, MIC determination, cytotoxicity/ADMET) is required before any conclusion about efficacy.

Top 10 Candidates for Experimental Testing

Ranked by CNS MPO score (5-parameter approximation).



Rank #1

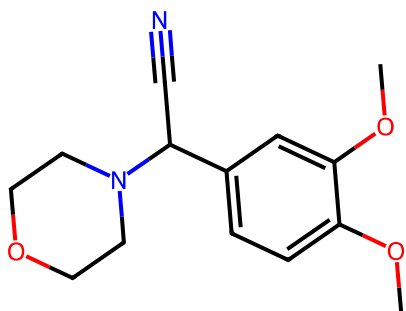
CNS MPO score: 5.000

QED: 0.799

MW: 300.4

Novel vs. training set: True

SMILES: C#CCN(C)C1CCc2ccc(OC(=O)N(C)C)cc2C1=O



Rank #2

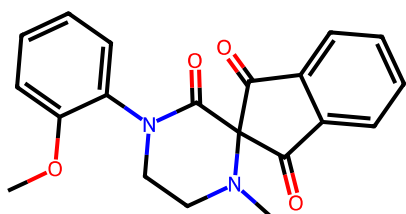
CNS MPO score: 5.000

QED: 0.825

MW: 262.3

Novel vs. training set: False

SMILES: COc1ccc(C(C#N)N2CCOCC2)cc1OC



Rank #3

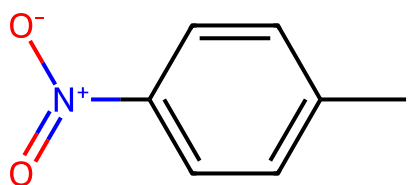
CNS MPO score: 5.000

QED: 0.773

MW: 350.4

Novel vs. training set: True

SMILES: COc1ccccc1N1CCN(C)C2(C(=O)c3ccccc3C2=O)C1=O



Rank #4

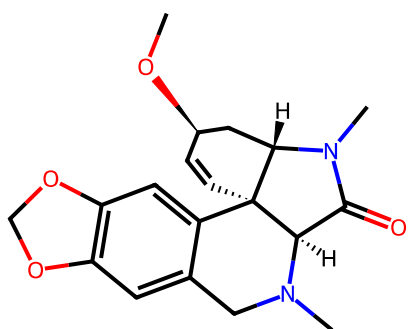
CNS MPO score: 5.000

QED: 0.438

MW: 137.1

Novel vs. training set: False

SMILES: Cc1ccc([N+](=O)[O-])cc1



Rank #5

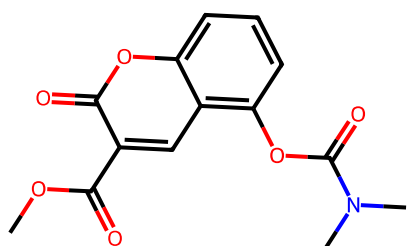
CNS MPO score: 5.000

QED: 0.720

MW: 342.4

Novel vs. training set: False

SMILES: CO[C@@H]1C=C[C@@]23c4cc5c(cc4CN(C)[C@@H]2C(=O)N(C)[C@H]3C1)OCO5



Rank #6

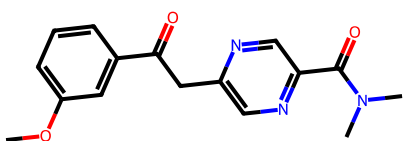
CNS MPO score: 5.000

QED: 0.617

MW: 291.3

Novel vs. training set: True

SMILES: COC(=O)c1cc2c(OC(=O)N(C)C)cccc2oc1=O



Rank #7

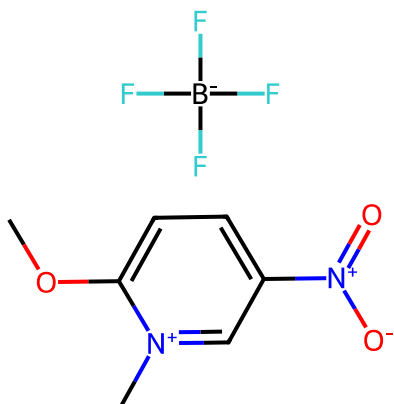
CNS MPO score: 5.000

QED: 0.784

MW: 299.3

Novel vs. training set: True

SMILES: COc1ccc(C(=O)Cc2cnc(C(=O)N(C)C)cn2)c1



Rank #8

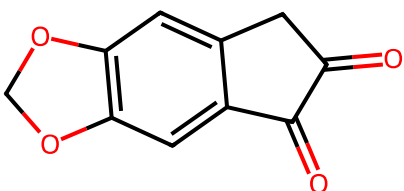
CNS MPO score: 5.000

QED: 0.266

MW: 256.0

Novel vs. training set: False

SMILES: COc1ccc([N+](=O)[O-])c[n+](C)1C.[B-](F)(F)F



Rank #9

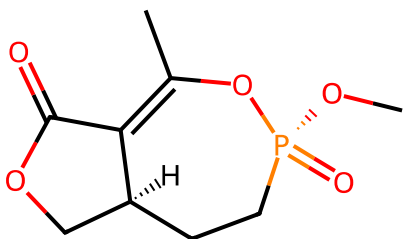
CNS MPO score: 5.000

QED: 0.566

MW: 190.2

Novel vs. training set: True

SMILES: O=C1Cc2cc3c(cc2C1=O)OCO3



Rank #10

CNS MPO score: 5.000

QED: 0.508

MW: 232.2

Novel vs. training set: False

SMILES: CO[P@]1(=O)CC[C@H]2COC(=O)C2=C(C)O1